Sindh Science Learning Portal

Class 5 · Science

Chapter 3

Seeds: Structure and Germination



Learn the parts of a seed, how a bean seed and a maize seed differ, and what a seed needs to germinate.

Following the Sindh Textbook Board curriculum

sindh-science-portal.vercel.app

CHAPTER 3

Seeds: Structure and Germination

What we will learn Learn the parts of a seed, how a bean seed and a maize seed differ, and what a seed needs to germinate.

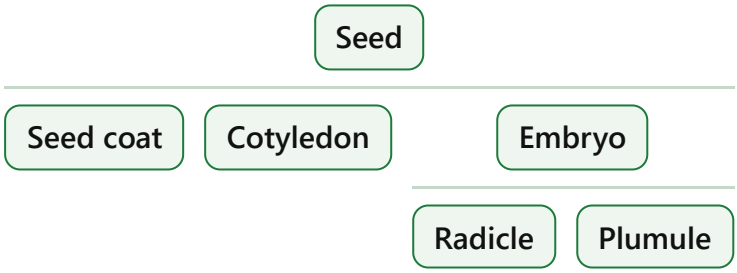
1 Parts of a seed

A new plant grows from a seed. Inside every seed there is a tiny plant and food for it.

- **Seed coat**: covers the seed and protects it. It has a tiny hole called the **micropyle**. Water goes in through it.
- **Embryo**: the baby plant inside the seed. The part that becomes the **root** is called the **radicle**, and the part that becomes the **stem and leaves** is called the **plumule**.
- **Cotyledons (seed leaves)**: they store **food** for the baby plant.

Seeds with two cotyledons are called **dicot seeds**, like beans, peas and chickpeas. Seeds with one cotyledon are called **monocot seeds**, like maize, wheat and rice.

Diagram: the parts and what they do



Seed (بِج · بیج)
A new plant grows from a seed. It holds a tiny plant and its food.

Seed coat (بِج جو کل · بیج کا چھلکا)
Protects the seed. Water enters through its tiny hole, the micropyle.

Cotyledon (بِج پن · دال پتا)
Stores food for the baby plant. A bean has two; maize has one.

Embryo (قوت · جنین)
The baby plant inside the seed, which grows into a new plant.

Radicle (ریڈیکل · ریڈیکل)
Comes out first and grows down to become the root.

Plumule (پلیومول · پلیومیول)
Grows up towards the light to become the stem and leaves.

2 A bean seed and a maize seed

- A **bean seed** is a dicot. Take the coat off a soaked bean and it opens easily into two halves. The two halves are the two **cotyledons**, and the tiny **embryo** is attached between them.
- A **maize grain** is a monocot. It has one cotyledon, an embryo and a large store of food called the **endosperm**.

In a bean, food is stored in the cotyledons. In maize, most of the food is stored in the endosperm.

We eat many seeds as food, such as lentils, chickpeas, peanuts and maize.

3 How does a seed germinate?

When a tiny plant comes out of a seed and starts to grow, we call it **germination**. The young new plant is called a **seedling**.

First the **radicle** comes out and grows down into the soil to become the root. Then the **plumule** grows up towards the light.

- **In a bean**: the stem bends like a hook, pushes out of the soil and pulls the cotyledons **above the soil**. The cotyledons turn green and make food for a few days, then dry up and fall off.
- **In maize**: the cotyledon stays **under the soil**. The soft plumule grows straight up inside a protective sheath.



Picture: Judgefloro · CC0 · Wikimedia Commons

4 What seeds need to germinate

Not every seed germinates. A seed needs three things to germinate:

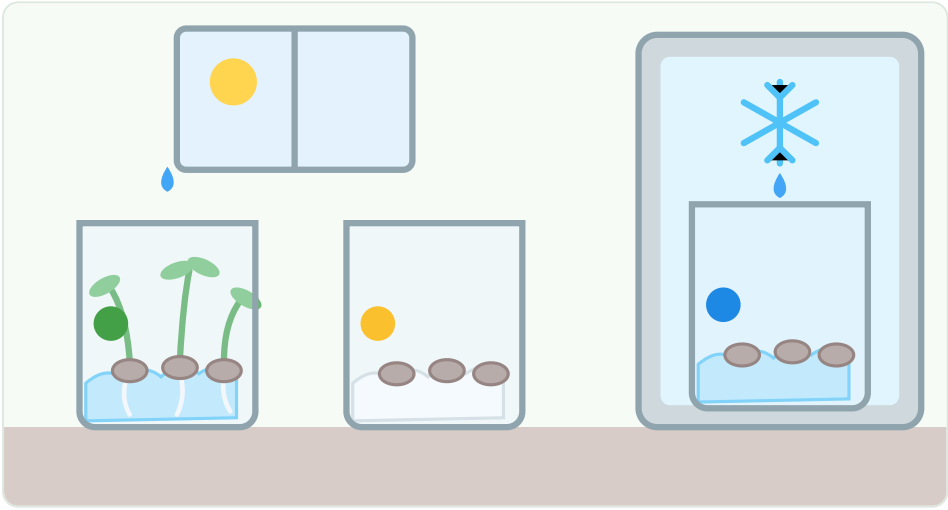
- **Water**: the seed soaks up water and swells, and its coat softens and splits. A dry seed does not germinate.
- **Air (oxygen)**: the seed has to breathe to get energy.
- **The right temperature**: most seeds germinate when it is warm. They do not germinate when it is very cold or very hot.

Some seeds also need light, while others germinate in the dark. A seed **does not need soil** to germinate: it can sprout on wet cotton. But to keep growing, the plant needs soil, sunlight and food.

Key words

English	سنڌي	اردو
seed	ٻج	بیج
seed coat	ٻج جو کل	بیج کا چھلکا
micropyle	مائڪروپائل	مائیکروپائل
embryo	ڏوٽ	جنین
radicle	ريڊيڪل	ریڈیکل
plumule	پليومول	پلیومیول
cotyledon	ٻج پن	دال پتا
endosperm	اينڈواسپرم	اینڈوسپرم
germination	ڦٽڻ	اگنا
seedling	ٻوٽڙو	پنیری

Experiment: What does a seed need to germinate?



You need

- Three small jars or cups
- Cotton wool
- 12 bean or chickpea seeds
- Water and labels

Steps

1. Label the jars A, B and C, and put cotton wool and 4 seeds in each.
2. Make the cotton in jar A wet and keep it in the room.
3. Leave the cotton in jar B dry and keep it in the room too.
4. Make the cotton in jar C wet and put it in the fridge.
5. Check every day for 5 days and keep A and C wet. Write down which seeds grew roots.

Safety: Ask an adult at home before using the fridge. Do not eat the seeds, and wash your hands afterwards.

What did we learn? Only the seeds in jar A germinated, where they had water, air and warmth. Jar B had no water and jar C was too cold. So a seed needs water, air and the right temperature to germinate.

Practice questions

Circle the correct answer

1. **Which part of a seed protects it?**

   (a) Seed coat **(b)** Radicle
   (c) Plumule **(d)** Endosperm

2. **Which part of the embryo becomes the root?**

   (a) Plumule **(b)** Radicle
   (c) Cotyledon **(d)** Micropyle

3. **Which of these is a dicot seed?**

   (a) Maize **(b)** Wheat
   (c) Bean **(d)** Rice

4. **Where is most of the food stored in a maize grain?**

   (a) In the endosperm **(b)** In the radicle
   (c) In the seed coat **(d)** In the micropyle

5. **How many cotyledons does a bean seed have?**

   (a) One **(b)** Two
   (c) Three **(d)** None

6. **What comes out of the seed first during germination?**

   (a) Leaves **(b)** A flower
   (c) The radicle **(d)** A fruit

7. **In which seed do the cotyledons come above the soil?**

   (a) Bean **(b)** Maize
   (c) Wheat **(d)** None of them

8. **Which of these does a seed NOT need to germinate?**

   (a) Water **(b)** Air
   (c) The right temperature **(d)** Soil

9. **Why does a seed need air (oxygen)?**

(a) To breathe and get energy (b) To change colour

(c) To become heavy (d) To dry out

Write true or false

10. **Cotyledons store food for the baby plant.** _____

11. **A maize grain has two cotyledons.** _____

12. **In maize, the cotyledon stays under the soil.** _____

13. **A dry seed can germinate without water.** _____

14. **A seed can germinate on wet cotton.** _____

Match the columns

15. Match each part of a seed with its job.

1. Radicle		(a) Becomes the stem and leaves
2. Plumule		(b) Lets water in
3. Micropyle		(c) Becomes the root

Match the columns

16. Match each seed with its type.

1. Pea		(a) Monocot seed
2. Wheat		(b) Dicot seed

Worksheet: Seeds: Structure and Germination

Name: _____ Date: _____

A. Fill in the blanks

Word box: endosperm · germination · plumule · seed coat · temperature

1. The cover on the outside of a seed is called the _____ .
2. The _____ of the embryo grows up into the stem and leaves.
3. In maize, food is stored in the _____ .
4. A tiny plant coming out of a seed and growing is called _____ .
5. A seed needs water, air and the right _____ to germinate.

B. Put ✓ for true and ✗ for false

1. ☐ A bean seed is a dicot seed.
2. ☐ The plumule comes out first during germination.
3. ☐ A seed must have soil to germinate.
4. ☐ A seed takes in water through the micropyle.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Radicle	( ) Monocot seed
2. Plumule	( ) Dicot seed
3. Maize	( ) Root
4. Chickpea	( ) Stem and leaves

D. Answer in short

1. Name the three main parts of a seed.

2. Write one difference between a bean seed and a maize grain.

3. What three things does a seed need to germinate?

E. Draw and label

Draw an opened bean seed and label the seed coat, cotyledons, radicle and plumule.

FOR TEACHERS

Answer key

Chapter 3: Seeds: Structure and Germination

Practice questions

1. (a) Seed coat
2. (b) Radicle
3. (c) Bean
4. (a) In the endosperm
5. (b) Two
6. (c) The radicle
7. (a) Bean
8. (d) Soil
9. (a) To breathe and get energy
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. seed coat · 2. plumule · 3. endosperm · 4. germination · 5. temperature

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✗ · 4. ✓

C. Match column A with column B (write the number in the brackets): (3) Monocot seed · (4) Dicot seed · (1) Root · (2) Stem and leaves

D. Answer in short:

1. Seed coat, embryo and cotyledons.
2. A bean has two cotyledons; maize has one cotyledon and an endosperm.
3. Water, air (oxygen) and the right temperature.

Picture credits

- Chapter 3: Judgefloro · CC0 · commons.wikimedia.org/wiki/File:1876Germination_of_Adzuki_bean_19.jpg

This book was made on the Sindh Science Learning Portal. Free and without ads. sindh-science-portal.vercel.app